

Лабораторная работа № 8

Настройка SMTP-сервера

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Раздел 1

Информация

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Раздел 2

Цель и задачи

- Получить навыки установки и базовой настройки SMTP-сервера Postfix.

- Установлены Postfix и s-nail, в firewalld открыт SMTP.

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- Ошибочная попытка доставки на client показала отсутствие SMTP-службы на клиенте; очередь была очищена.

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- Настройки перенесены в server-mail и client-mail provisioners.

Раздел 3

Ход работы

Установка Postfix

- На сервер установлены Postfix и консольный почтовый клиент s-nail; SMTP разрешён в firewall.

```
root@server:~#
(1/2): postfix-lmdb-3.8.5-10.el10_2.x86_64.rpm      188 kB/s | 23 kB    00:00
(2/2): postfix-3.8.5-10.el10_2.x86_64.rpm          4.6 MB/s | 1.5 MB   00:00
-----
Total                                              2.4 MB/s | 1.5 MB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : postfix-lmdb-2:3.8.5-10.el10_2.x86_64      1/
  Installing               : postfix-2:3.8.5-10.el10_2.x86_64          1/
  Running scriptlet: postfix-2:3.8.5-10.el10_2.x86_64                  2/
  Installing              : postfix-2:3.8.5-10.el10_2.x86_64          2/
  Running scriptlet: postfix-2:3.8.5-10.el10_2.x86_64                  2/

Installed:
  postfix-2:3.8.5-10.el10_2.x86_64      postfix-lmdb-2:3.8.5-10.el10_2.x86_64

Complete!
Last metadata expiration check: 0:00:18 ago on Sun 06 Sep 2026 08:37:39 AM UTC.
Dependencies resolved.

=====
Package           Architecture Version           Repository      Size
=====
Installing:
s-nail            x86_64        14.9.24-12.el10 appstream       633

Transaction Summary
=====
Install 1 Package

Total download size: 633 k
Installed size: 1.2 M
Downloading Packages:
s-nail-14.9.24-12.el10.x86_64.rpm      3.2 MB/s | 633 kB    00:00
-----
Total                                  1.3 MB/s | 633 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
```

Имя почтового сервера

- В main.cf заданы myhostname и mydomain.

```
root@server:~  
root@server:~# firewall-cmd --add-service=smtp  
firewall-cmd --add-service=smtp --permanent  
firewall-cmd --list-services  
success  
success  
cockpit dhcp dhcpv6-client dns http https smtp ssh ssh-custom  
root@server:~# restorecon -vR /etc  
systemctl enable postfix  
systemctl start postfix  
systemctl status postfix --no-pager  
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t  
:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0  
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' -> '/usr/lib/systemd/system  
/postfix.service'.  
• postfix.service - Postfix Mail Transport Agent  
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: disabled)  
   Active: active (running) since Sun 2026-09-06 08:38:39 UTC; 35ms ago  
   Invocation: e4a2f40609374c13b3c452158bc14d68  
   Process: 12870 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (code=exited, status=0/SU  
CCESS)  
   Process: 12871 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)  
   Process: 12876 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)  
   Process: 12878 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)  
   Main PID: 12947 (master)  
   Tasks: 3 (limit: 10232)  
   Memory: 2.8M (peak: 3.4M)  
   CPU: 595ms  
   CGroup: /system.slice/postfix.service  
           └─12947 /usr/libexec/postfix/master -w  
             └─12948 pickup -l -t unix -u  
             └─12949 qmgr -l -t unix -u
```

Источник и локальные домены

- Параметры myorigin и mydestination настроены для домена ndbaranov.net.

```
root@server:~# postconf myorigin
myorigin = $myhostname
root@server:~# postconf mydomain
mydomain = localdomain
root@server:~# postconf -e 'myorigin = $mydomain'
root@server:~# postconf myorigin
myorigin = $mydomain
root@server:~# postfix check
root@server:~# systemctl reload postfix
root@server:~# postconf -n
alias_database = ldap:/etc/aliases
alias_maps = ldap:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 3.8
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name
$process_id & sleep 5
default_database_type = ldap
html_directory = no
inet_interfaces = localhost
inet_protocols = all
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
myorigin = $mydomain
newaliases_path = /usr/bin/newaliases.postfix
queue_directory = /var/spool/postfix
readme_directory = /usr/share/doc/postfix/README_FILES
sample_directory = /usr/share/doc/postfix/samples
sendmail_path = /usr/sbin/sendmail.postfix
setgid_group = postdrop
shlib_directory = /usr/lib64/postfix
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
smtp_tls_CApith = /etc/pki/tls/certs
smtp_tls_security_level = may
smtpd_tls_cert_file = /etc/pki/tls/certs/postfix.pem
smtpd_tls_key_file = /etc/pki/tls/private/postfix.key
smtpd_tls_security_level = may
unknown_local_recipient_reject_code = 550
root@server:~# postconf -e 'mydomain = ndbaranov.net'
> exit
```

Сетевые интерфейсы Postfix

- Сервер переведён на IPv4 и настроен на приём соединений с внутренних интерфейсов.

```
root@server:~  
root@server:~# postconf myhostname  
postconf mydomain  
postconf myorigin  
postconf inet_protocols  
myhostname = server.ndbaranov.net  
mydomain = ndbaranov.net  
myorigin = $mydomain  
inet_protocols = ipv4  
root@server:~#
```

Проверка конфигурации

- Команда postfix check завершилась без сообщений об ошибках, после чего служба перезапущена.

```
Выбрать root@server:~
PS D:\work\ndbaranov\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 08:37:03 2026 from 10.0.2.2
vagrant@server:~$ sudo -i
root@server:~# tail -f /var/log/maillog
Sep  6 08:43:36 server postfix/postfix-script[13186]: refreshing the Postfix mail system
Sep  6 08:43:36 server postfix/master[12947]: reload -- version 3.8.5, configuration /etc/postfix
Sep  6 08:43:36 server postfix/master[12947]: warning: ignoring inet_protocols parameter value change
Sep  6 08:43:36 server postfix/master[12947]: warning: old value: "all", new value: "ipv4"
Sep  6 08:43:36 server postfix/master[12947]: warning: to change inet_protocols, stop and start Postfix
Sep  6 08:44:59 server postfix/pickup[13197]: 6184D60D7EAA: uid=1001 from=<ndbaranov>
Sep  6 08:44:59 server postfix/cleanup[13268]: 6184D60D7EAA: message-id=<20260906084459.6184D60D7EAA@server.ndbaranov.net>
Sep  6 08:44:59 server postfix/qmgr[13198]: 6184D60D7EAA: from=<ndbaranov@ndbaranov.net>, size=340, nrcpt=1 (queue active)
Sep  6 08:44:59 server postfix/local[13270]: 6184D60D7EAA: to=<ndbaranov@server.ndbaranov.net>, relay=local, delay=0.08, delays=0.06/0.02/0/0.01, dsn=2.0.0, status=sent (delivered to mailbox)
Sep  6 08:44:59 server postfix/qmgr[13198]: 6184D60D7EAA: removed
```

Отправка локального письма

- Через s-nail отправлено тестовое письмо локальному пользователю.

```
root@client-  
Total download size: 1.5 M  
Installed size: 4.5 M  
Downloading Packages:  
(1/2): postfix-lmdb-3.8.5-10.el10_2.x86_64.rpm      175 kB/s | 23 kB  00:00  
(2/2): postfix-3.8.5-10.el10_2.x86_64.rpm          3.9 MB/s | 1.5 MB 00:00  
-----  
Total                                              2.3 MB/s | 1.5 MB 00:00  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing      :                                1/1  
  Installing     : postfix-lmdb-2:3.8.5-10.el10_2.x86_64 1/2  
  Running scriptlet: postfix-2:3.8.5-10.el10_2.x86_64 2/2  
  Installing     : postfix-2:3.8.5-10.el10_2.x86_64 2/2  
  Running scriptlet: postfix-2:3.8.5-10.el10_2.x86_64 2/2  
  
Installed:  
postfix-2:3.8.5-10.el10_2.x86_64      postfix-lmdb-2:3.8.5-10.el10_2.x86_64  
  
Complete!  
Last metadata expiration check: 0:00:14 ago on Sun 06 Sep 2026 08:51:19 AM UTC.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing: s-nail	x86_64	14.9.24-12.el10	appstream	633 k

```
=====
```

=====				
Install 1 Package				
Total download size: 633 k				
Installed size: 1.2 M				
Downloading Packages:				
s-nail-14.9.24-12.el10.x86_64.rpm		461 kB/s 633 kB		00:01

Total		359 kB/s 633 kB		00:01

```
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction
```

Запись о локальной доставке

- В maillog видна цепочка обработки письма и статус delivered/sent.

```
root@server:~  
ou have new mail in /var/spool/mail/ndbaranov  
ndbaranov@server ~]$ sudo -i  
sudo] password for ndbaranov:  
root@server ~]# tail -f /var/log/maillog  
ep 6 08:43:36 server postfix/postfix-script[13186]: refreshing the Postfix mail system  
ep 6 08:43:36 server postfix/master[12947]: reload -- version 3.8.5, configuration /etc/postfix  
ep 6 08:43:36 server postfix/master[12947]: warning: ignoring inet_protocols parameter value change  
ep 6 08:43:36 server postfix/master[12947]: warning: old value: "all", new value: "ipv4"  
ep 6 08:43:36 server postfix/master[12947]: warning: to change inet_protocols, stop and start Postfix  
ep 6 08:44:59 server postfix/pickup[13197]: 6184D60D7EAA: uid=1001 from=<ndbaranov>  
ep 6 08:44:59 server postfix/cleanup[13268]: 6184D60D7EAA: message-id=<20260906084459.6184D60D7EAA@server.ndbaranov.net>  
ep 6 08:44:59 server postfix/qmgr[13198]: 6184D60D7EAA: from=<ndbaranov@ndbaranov.net>, size=340, nrcp=1 (queue active)  
ep 6 08:44:59 server postfix/local[13270]: 6184D60D7EAA: to=<ndbaranov@server.ndbaranov.net>, relay=local, delay=0.08, delays=0.06/0.02/0/0.01, dsn=2.0.0, status=sent (delivered to mailbox)  
ep 6 08:44:59 server postfix/qmgr[13198]: 6184D60D7EAA: removed
```

Приём SMTP из внутренней сети

- В mynetworks добавлена лабораторная подсеть.

```
root@server:~  
[root@server ~]# postconf inet_interfaces  
postconf mynetworks  
inet_interfaces = localhost  
mynetworks = 127.0.0.1/32  
[root@server ~]# postconf -e 'inet_interfaces = all'  
[root@server ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'  
[root@server ~]# postfix check  
systemctl reload postfix  
systemctl stop postfix  
systemctl start postfix  
[root@server ~]# grep -n "Subject: test-client-after" /var/spool/mail/ndbaranov  
[root@server ~]# tail -f /var/log/maillog  
Sep  6 08:54:53 server postfix/master[12947]: terminating on signal 15  
Sep  6 08:54:54 server postfix/postfix-script[13706]: starting the Postfix mail system  
Sep  6 08:54:54 server postfix/master[13708]: daemon started -- version 3.8.5, configuration /etc/postfix  
x  
Sep  6 08:55:05 server postfix/smtpd[13714]: connect from client.ndbaranov.net[192.168.1.30]  
Sep  6 08:55:05 server postfix/smtpd[13714]: C5ABD60D7EA3: client=client.ndbaranov.net[192.168.1.30]  
Sep  6 08:55:05 server postfix/cleanup[13718]: C5ABD60D7EA3: message-id=<20260906085505.9F61EF8881@client.ndbaranov.net>  
Sep  6 08:55:05 server postfix/qmgr[13710]: C5ABD60D7EA3: from=<ndbaranov@client.ndbaranov.net>, size=573, nrcpt=1 (queue active)  
Sep  6 08:55:05 server postfix/smtpd[13714]: disconnect from client.ndbaranov.net[192.168.1.30] ehlo=2 s
```

Отправка письма с client

- С client отправлено письмо пользователю на server.ndbaranov.net.

```
root@server:~  
[root@server ~]# tail -f /var/log/maillog  
Sep  6 08:57:32 server postfix/cleanup[13757]: 6730060D7EA3: message-id=<20260906085732.49534F8881@client.ndbaranov.net>  
Sep  6 08:57:32 server postfix/qmgr[13710]: 6730060D7EA3: from=<ndbaranov@client.ndbaranov.net>, size=546, nrcpt=1 (queue active)  
Sep  6 08:57:32 server postfix/smtpd[13754]: disconnect from client.ndbaranov.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7  
Sep  6 08:57:32 server postfix/smtp[13760]: 6730060D7EA3: to=<ndbaranov@ndbaranov.net>, relay=none, delay=0.08, delays=0.03/0.05/0/0, dsn=5.4.6, status=bounced (mail for ndbaranov.net loops back to myself)  
Sep  6 08:57:32 server postfix/cleanup[13757]: 7AF0A60D7EA4: message-id=<20260906085732.7AF0A60D7EA4@server.ndbaranov.net>  
Sep  6 08:57:32 server postfix/bounce[13761]: 6730060D7EA3: sender non-delivery notification: 7AF0A60D7EA4  
Sep  6 08:57:32 server postfix/qmgr[13710]: 7AF0A60D7EA4: from=<>, size=2550, nrcpt=1 (queue active)  
Sep  6 08:57:32 server postfix/qmgr[13710]: 6730060D7EA3: removed  
Sep  6 08:57:32 server postfix/smtp[13760]: connect to client.ndbaranov.net[192.168.1.30]:25: No route to host  
Sep  6 08:57:32 server postfix/smtp[13760]: 7AF0A60D7EA4: to=<ndbaranov@client.ndbaranov.net>, relay=none, delay=0.02, delays=0.02/0/0/0, dsn=4.4.1, status=deferred (connect to client.ndbaranov.net[192.168.1.30]:25: No route to host)  
^C  
[root@server ~]# postqueue -p  
-Queue ID- --Size-- ----Arrival Time---- -Sender/Recipient-----  
7AF0A60D7EA4    2550 Sun Sep  6 08:57:32 MAILER-DAEMON  
(connect to client.ndbaranov.net[192.168.1.30]:25: No route to host)
```

Диагностика неуспешной доставки

- Попытка отправить письмо на узел без SMTP-службы оставила сообщение в очереди.

```
root@server:~# rndc freeze ndbaranov.net
rndc freeze 1.168.192.in-addr.arpa
[root@server ~]# nano /var/named/master/fz/ndbaranov.net
[root@server ~]# nano /var/named/master/fz/ndbaranov.net
[root@server ~]# nano /var/named/master/rz/192.168.1
[root@server ~]# rndc thaw ndbaranov.net
rndc thaw 1.168.192.in-addr.arpa
A zone reload and thaw was started.
Check the logs to see the result.
A zone reload and thaw was started.
Check the logs to see the result.
[root@server ~]# dig MX ndbaranov.net
dig mail.ndbaranov.net

<<>> DiG 9.18.33 <<>> MX ndbaranov.net
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 29943
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 7e6567566f5566d0010000006a9d2c4020b87be89d576af4 (good)
;; QUESTION SECTION:
; ndbaranov.net.                IN      MX

;; ANSWER SECTION:
ndbaranov.net.                86400   IN      MX      10 mail.ndbaranov.net.

;; ADDITIONAL SECTION:
mail.ndbaranov.net.          86400   IN      A        192.168.1.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Sun Sep 06 09:02:56 UTC 2026
;; MSG SIZE rcvd: 107

<<>> DiG 9.18.33 <<>> mail.ndbaranov.net
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 8538
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 3

;; OPT PSEUDOSECTION:
```

Очистка почтовой очереди

- После анализа ошибочное сообщение удалено из очереди.

```
Выбрать root@server:~
[root@server ~]# postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
[root@server ~]# postfix check
systemctl reload postfix
[root@server ~]# restorecon -vR /etc
restorecon -vR /var/named
[root@server ~]# systemctl restart named
[root@server ~]# postqueue -f
[root@server ~]# postqueue -p
-Queue ID- --Size-- ----Arrival Time---- -Sender/Recipient-----
7AF0A60D7EA4    2550 Sun Sep  6 08:57:32 MAILER-DAEMON
      (connect to client.ndbaranov.net[192.168.1.30]:25: No route to host)
                        ndbaranov@client.ndbaranov.net

-- 2 Kbytes in 1 Request.
[root@server ~]# grep -n "Subject: final-test" /var/spool/mail/ndbaranov
63:Subject: final-test
[root@server ~]#
```

Автоматизация почтовой настройки

- Provision-сценарии server-mail и client-mail устанавливают пакеты и применяют параметры Postfix.

```
PowerShell 7 (x64)
[ndbaranov@client.ndbaranov.net ~]$ echo . | mail -s final-test ndbaranov@ndbaranov.net
[ndbaranov@client.ndbaranov.net ~]$ exit
logout
vagrant@client:~$ exit
logout
PS D:\work\ndbaranov\vagrant> cd D:\work\ndbaranov\vagrant
PS D:\work\ndbaranov\vagrant> notepad .\provision\server\mail.sh
PS D:\work\ndbaranov\vagrant> notepad .\provision\client\mail.sh
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server mail"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server mail (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-22532-9o9doa.sh
server: Provisioning Postfix server
server: Last metadata expiration check: 0:17:09 ago on Sun 06 Sep 2026 08:51:38 AM UTC.
server: Package postfix-2:3.8.5-10.el10_2.x86_64 is already installed.
server: Dependencies resolved.
server: Nothing to do.
server: Complete!
server: Last metadata expiration check: 0:17:10 ago on Sun 06 Sep 2026 08:51:38 AM UTC.
server: Package s-nail-14.9.24-12.el10.x86_64 is already installed.
server: Dependencies resolved.
server: Nothing to do.
server: Complete!
server: Warning: ALREADY_ENABLED: smtp
server: success
server: success
server: Postfix server provisioning completed
PS D:\work\ndbaranov\vagrant> vagrant provision client --provision-with "client mail"
==> client: Running provisioner: client mail (shell)...
client: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-23420-3t38cr.sh
```

Раздел 4

Результаты

- Postfix принимает и доставляет локальную и сетевую почту. Анализ очереди и журнала позволил отличить успешную доставку от ошибки соединения с узлом без SMTP-службы.

Спасибо за внимание